

RESEARCH ARTICLE

Silk Fibroin-Regulated Shape Memory Composite Hydrogels with Tunable Recovery Time

 Zhengdong Feng¹  | Wei Huang² | Fei Lv¹ | Wei Li¹
¹College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing, P. R. China | ²Department of Mechanical Engineering, University College London, London, United Kingdom

Correspondence: Zhengdong Feng (zhengdong.feng@njupt.edu.cn) | Wei Li (wei.li@njupt.edu.cn)

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ABSTRACT

Shape memory polymers (SMPs) are pivotal for applications in soft robotics, biomedical devices, and flexible electronics. However, their reliance on external stimuli for activation and the difficulty of precisely programming recovery dynamics remain significant limitations. Autonomous SMPs with delayed recovery offer a promising alternative, yet facile control of recovery onset and duration is still difficult to attain. Here, a semi-interpenetrating network (semi-IPN) hydrogel composed of poly(acrylic acid) and silk fibroin (PAA-SF) is presented, fabricated via a facile one-step UV polymerization. By modulating the SF content within the PAA network, the hydrogel exhibits tunable shape recovery kinetics across multiple programming environments, including organic solvents, air, and water, enabling spatiotemporal control over shape restoration. Moreover, the hydrogels show significantly enhanced mechanical properties and strong adhesion compared to neat PAA hydrogels. Benefiting from these combined properties and excellent cytocompatibility, demonstrating two proof-of-concept devices: a smart four-jaw gripper capable of retrieving objects in confined spaces, and a stiffness-switchable needle that reversibly transitions between soft and rigid states. This work not only introduces a robust and multifunctional shape memory hydrogel but also provides a versatile platform for designing stimulus-free, time-programmable soft actuators for biomedical and robotic applications.

1 | Introduction

Shape memory polymers (SMPs) constitute a class of smart materials that can be programmed into a temporary shape at a specific temperature and subsequently recover to their original, permanent shape upon exposure to an external stimulus [1, 2]. In recent decades, stimulus-responsive SMPs have garnered extensive attention due to their broad potential in emerging technologies such as soft robotics, minimally invasive medical devices, flexible electronics, and aerospace systems [3–6].

Nevertheless, critical challenges remain regarding both the recovery dynamics and the type of stimulus required [4, 5]. Thermal activation is the most widely used trigger for SMPs [17], yet it is often impractical in constrained or sensitive environments, such as deep tissues in the human body or extreme aerospace conditions [6, 7]. To address these limitations, a variety of alternative stimuli have been explored, including light [8], magnetic fields [9], and electricity [10]. However, each approach presents inherent limitations [11]; for instance, light suffers from limited penetration depth in turbid media [12], while electrical actuation typically requires wired connections and complex circuitry

Zhengdong Feng and Wei Huang contributed equally to this work.

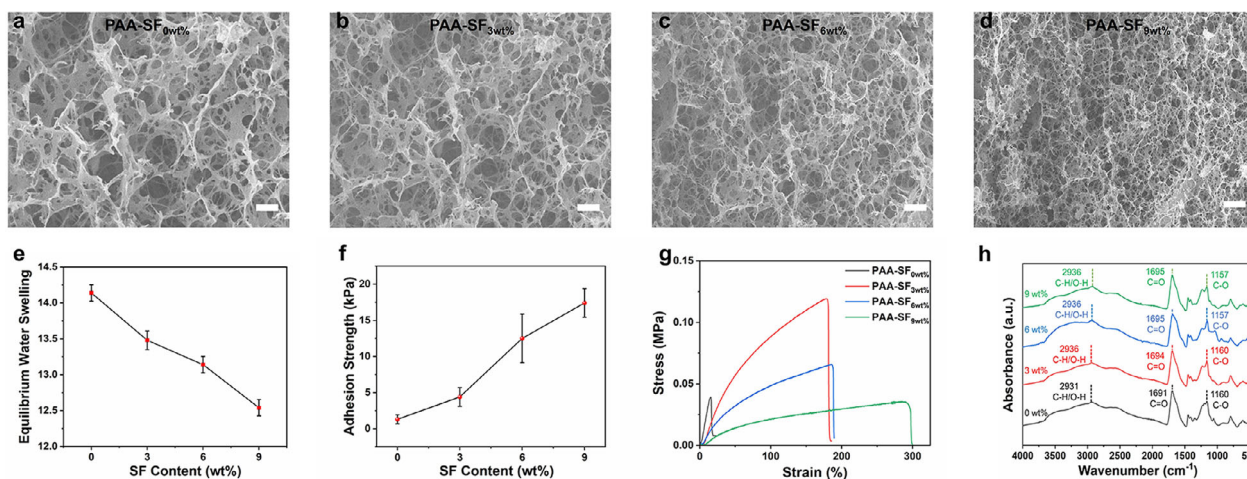


FIGURE 1 | (a–d) SEM images of the cross-sectional morphology of samples with various SF content. All scale bars are 5 μm. (e) Equilibrium swelling ratio of PAA-SF hydrogels. (f) The lap-shear tests of PAA-SF hydrogel adhesion at room temperature and 40% relative humidity. (g) Tensile stress–strain curves of PAA-SF hydrogels. (h) FTIR spectra of PAA-SF xerogel powders with different SF contents. All the error bars correspond to s.d. ($n = 5$).

in hydrogen bonds within the network, thereby reinforcing interfacial adhesion [33]. Tensile tests were also conducted to reveal the mechanical impact of SF incorporation. As shown in Figure 1g, the fracture elongation increases dramatically with SF content; in particular, the PAA-SF_{9 wt.%} hydrogel exhibits an elongation at break of ~300%, more than 15 times that of the neat PAA hydrogel. Conversely, the tensile strength and Young's modulus (Figure S2) decrease when the SF content exceeds 3 wt.%, likely due to the disruption of the continuous PAA network by excessive SF domains [18].

Fourier transform infrared spectroscopy (FTIR) was employed to verify the molecular interactions within the hydrogel network. The spectra of PAA xerogel powders with different SF contents are presented in Figure 1h. The pristine PAA hydrogel (0 wt.% SF) shows characteristic peaks at 1691 cm⁻¹ (C=O stretching of carboxyl groups) and a broad band around 2931 cm⁻¹ (C–H stretching overlapped with hydrogen-bonded O–H). Upon incorporation of SF, systematic changes are observed: the C=O absorption peak at 1691 cm⁻¹ shifts in position and intensity, and the O–H related bands broaden. These spectral changes confirm the formation of intermolecular hydrogen bonds between functional groups (N–H, O–H) of the SF and the C=O groups of PAA chains. The most pronounced peak shift and intensity change at 1157 cm⁻¹ (C–O stretching) occur at the 6 wt.% SF, suggesting an optimal crosslinking density and the most complete formation of hydrogen bonds and other physical linkages at this composition [18].

The opposing thermal responses of the PAA-SF hydrogels were elucidated by variable temperature Fourier Transform Infrared Spectroscopy (VT-FTIR) (Figure 2a,b; Figure S3). A consistent blue shift of the band near 1750 cm⁻¹ was observed for both 0 and 6 wt.% SF samples upon heating, indicative of dehydration and strengthening of the PAA network [34]. Concurrently, a distinct blue shift near 1610 cm⁻¹ in the 6 wt.% SF sample confirms temperature-driven formation and consolidation of SF β-sheets [35]. During hydration-mediated recovery at room temperature, both peaks exhibit a red shift. The reversal at ~1750 cm⁻¹ reflects

rehydration and ionization of PAA [36], whereas the red shift at ~1610 cm⁻¹ provides molecular evidence of the disassembly of the SF β-sheet physical crosslinks [37, 38]. These complementary spectral shifts corroborate a cooperative mechanism in which SF conformational switching and PAA network dynamics jointly govern the shape memory cycle. The Raman spectra of PAA-SF_{0 wt.%} and PAA-SF_{6 wt.%} hydrogels in water and IPA further reveal distinct molecular mechanisms (Figure 2c). In water, both samples show a peak near 1410 cm⁻¹, attributed to the symmetric stretching of PAA carboxylate ions (–COO⁻) [39], suggesting that PAA ionization drives hydrogel swelling through electrostatic repulsion and disrupts physical crosslinks, thereby powering shape recovery [22]. In IPA, the 6 wt.% SF sample exhibits characteristic peaks at 1348.6 cm⁻¹ and 495.87 cm⁻¹. The former can be associated with the amide III band, signifying SF β-sheet formation and evidencing IPA-induced conformational fixation [40, 41], while the latter arises from skeletal/S–S vibrations, collectively confirming SF's structural role [42]. In contrast, the 0 wt.% sample shows a peak at 1139.27 cm⁻¹, corresponding to C–O stretching associated with hydrogen bonding between PAA and IPA [43], illustrating a fundamentally distinct, SF-independent shape-fixing mechanism [44]. These findings provide molecular-level validation of the “hydration-triggered recovery” and “solvent-programmed fixation” behaviors.

SEM was further employed to correlate morphology with performance after programming under different conditions. Figure 3a–c shows the surface morphology of PAA-SF_{6 wt.%} hydrogels programmed in 20°C IPA, 80°C air, and 80°C water, respectively. The sample programmed in 80°C air exhibits the largest surface pores, indicating the most pronounced phase separation. This morphology directly corresponds to its longest recovery time (t_{recovery}) (Figure 3k), since the highly porous structure hinders rapid water ingress and network plasticization. Figure 3d–f presents the shape recovery kinetics of the PAA-SF_{6 wt.%} hydrogel after being programmed in 20°C IPA, 80°C air, and 80°C water for 5 min, respectively. A distinct recovery delay period is observed at the initial stage, during which little macroscopic recovery occurs, attributed to a ‘recovery readiness’ phase required for

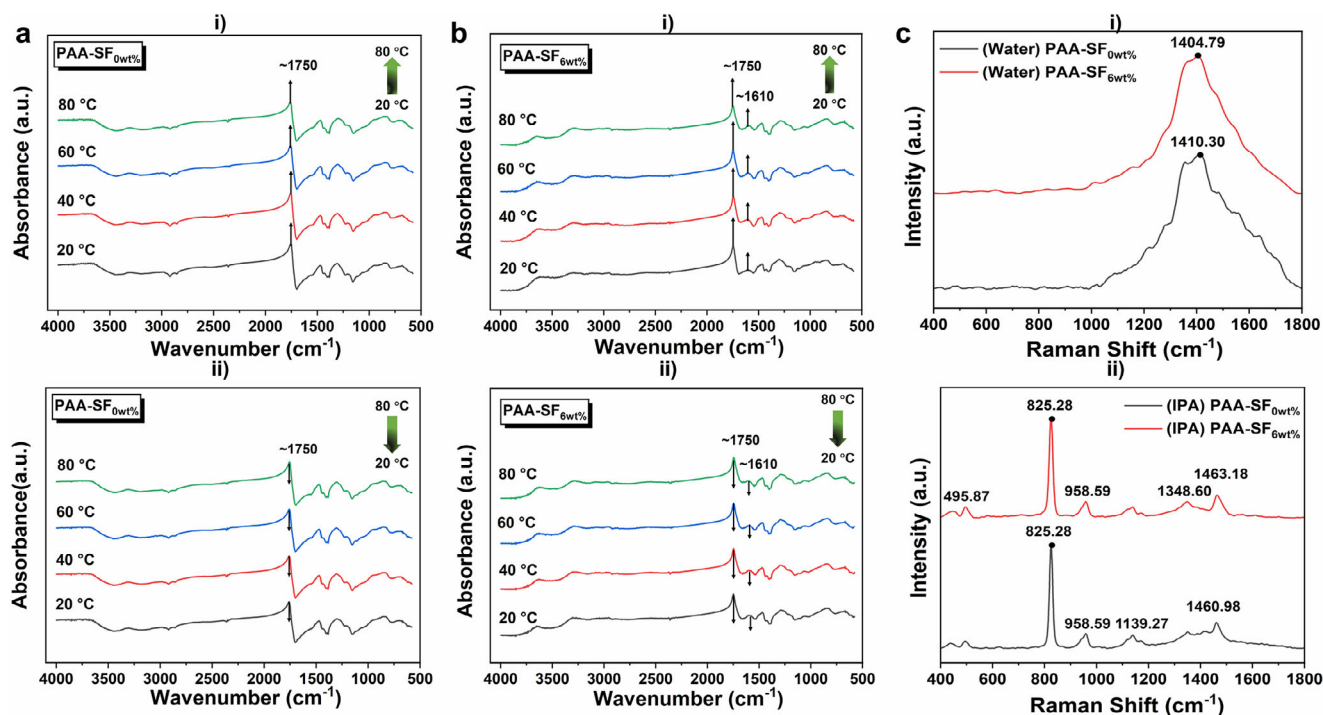


FIGURE 2 | (a) VT-FTIR spectra of PAA-SF xerogel powder with 0 wt.% SF acquired during i) the programming process and ii) the recovery process (total duration: 20 min). (b) VT-FTIR spectra of PAA-SF xerogel powder with 6 wt.% SF acquired during i) the programming process and ii) the recovery process (total duration: 20 min). (c) Raman spectra of PAA-SF₀ wt.% and PAA-SF₆ wt.% hydrogels i) in water and ii) in IPA.

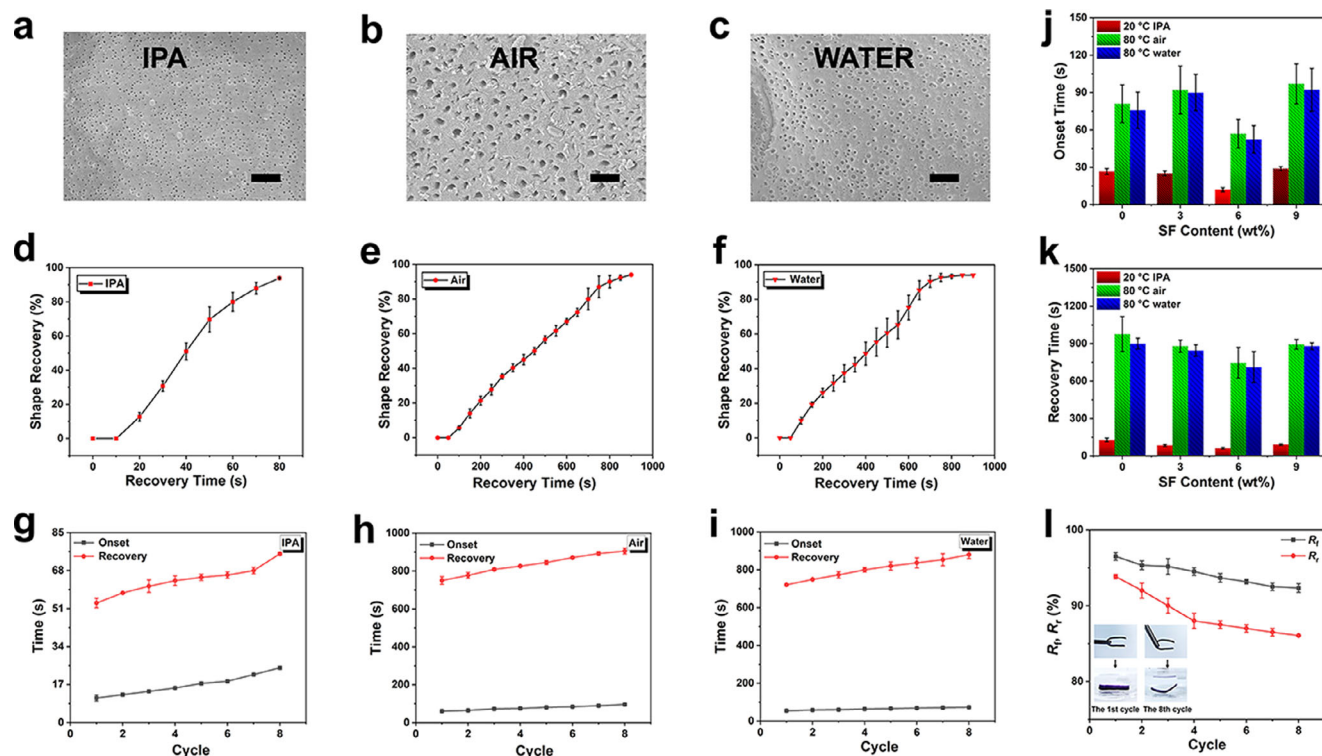


FIGURE 3 | (a–c) SEM images of the surface morphology of PAA-SF₆ wt.% hydrogels programmed in 20°C IPA, 80°C air, and 80°C water for 5 min, respectively. All scale bars are 5 μ m. (d–f) Shape recovery kinetics of PAA-SF₆ wt.% hydrogels programmed in 20°C IPA, 80°C air, and 80°C water for 5 min, respectively. (g–i) Onset time (t_{onset}) and recovery time (t_{recovery}) of PAA-SF₆ wt.% hydrogel programmed in 20°C IPA, 80°C air, and 80°C water for 5 min, respectively (8 cycles). (j) t_{onset} and (k) t_{recovery} of PAA-SF hydrogel programmed in 20°C IPA, 80°C air, and 80°C water for 5 min. (l) Shape fixity (R_f) and shape recovery (R_r) ratios of the PAA-SF₆ wt.% hydrogels across 8 programming/recovery cycles, via immersion in 20°C IPA for 5 min per cycle. All the error bars correspond to s.d. ($n = 5$).

network disassociation and rehydration. This trend aligns well with the onset time (t_{onset}) shown in Figure 3j. Following this period, the hydrogel gradually reverts to its original shape. As hydration proceeds, the PAA-SF_{6 wt.%} hydrogel ultimately achieves a high recovery extent exceeding 90%. The cyclic durability of the PAA-SF_{6 wt.%} hydrogel was evaluated over multiple programming/recovery cycles (Figure 3g–i). Hydrogel strips were fixed in U-shape (Figure S4). In all conditions, recovery time progressively increased with cycle number, which is attributed to the accumulation of residual stress within the hydrogel network under repeated deformation. Figure 3j,k summarizes t_{onset} and t_{recovery} for hydrogels with varying SF content programmed in three environments: 20°C IPA, 80°C air, and 80°C water. The PAA-SF_{6 wt.%} hydrogel consistently exhibits the fastest recovery kinetics across all conditions. We propose that SF modulates the effective crosslinking density, thereby increasing hydrogel stiffness and promoting faster recovery. However, optimal SF content exists at 6 wt.%; further increasing SF content (e.g., 9 wt.%) leads to an overly complex internal structure that begins to impede recovery. Notably, programming in 80°C air results in the longest t_{onset} and t_{recovery} . This is likely because the relatively dry state during air programming promotes a more pronounced phase separation (as confirmed by SEM in Figure 3b), creating a tortuous path for water migration upon immersion and thus delaying hydration and chain reconfiguration. Furthermore, despite a decline after 8 cycles (Figure 3l), the shape fixity ratio (R_f) and shape recovery ratio (R_r) remained above 90% and 85%, respectively, underscoring the excellent repeatability of the hydrogels.

All PAA-SF hydrogels exhibit excellent shape memory performance, with both R_f and R_r surpassing 90% in all three programming environments (Figure 4a–c). The swelling behavior after programming reveals a consistent inverse correlation between SF content and swelling ratio across all programming environments (Figure 4d–f), in agreement with equilibrium swelling data (Figure 1e). The initial swelling [swelling ratio after programming (pre-hydration)] state significantly varied with programming conditions: hydrogels programmed in 80°C air show the lowest initial swelling, indicating strong dehydration during programming, whereas those programmed in 80°C water exhibit relatively rapid initial swelling. Samples programmed in 20°C IPA maintain a moderate initial swelling level, suggesting partial solvent retention. Swelling kinetics further elucidate the shape memory mechanism. All samples exhibit an initial slow-swelling phase—most evident in air-programmed hydrogels—corresponding to the onset time (t_{onset}). This was followed by rapid swelling once continuous hydration pathways formed. Swelling rates also varied with programming environment: IPA-programmed hydrogels swell fastest due to their more homogeneous network; air-programmed hydrogels swell slowest due to the most extensive phase separation; and water-programmed hydrogels exhibit an intermediate rate. These findings directly link swelling behavior to shape recovery performance, highlighting the crucial roles of programming conditions.

The remarkable shape memory capability is visually demonstrated in Figure 4g (Figure S5 and Movies S1 and S2). A 3D-printed flower-shaped hydrogel (PAA-SF_{6 wt.%}) was programmed into a “flower bud” temporary shape by fixation in 20°C IPA for 5 min. Upon immersion in 20°C water, the hydrogel under-

went hydration-mediated recovery and fully bloomed back to its original shape within 10 min 33 s. A critical onset period of 0–1 min is observed, corresponding to the time required to dissociate temporary physical crosslinks and attain sufficient chain mobility for network reconfiguration.

The utility of the PAA-SF hydrogels for soft robotic gripping is demonstrated using a 3D-printed four-jaw gripper (Figure 5). Figure 5a shows a gripper whose original configuration is closed. The left panel shows the gripper in its programmed temporary shape, while the right panel depicts its return to the original closed state. Figure 5b (Movie S3) captures the gripper approaching an object (left) and successfully grasping it (right), demonstrating its capability to retrieve items from confined spaces. A complementary design with an originally open configuration is shown in Figure 5c. The left image shows the programmed shape, and the right image displays the restored open state after shape recovery. Figure 5d (Movie S4) documents the gripper approaching an object prior to release (left) and after complete release (right). Together, these demonstrations indicate that the PAA-SF_{6 wt.%} gripper design is well-suited for transporting objects within constrained environments, where autonomous, time-programmed recovery can be exploited.

A more advanced demonstration is a stiffness-tunable hydrogel needle, as shown in Figure 6a. The PAA-SF_{6 wt.%} needle can be mounted to a standard syringe (Figure 6b; Figure S6). After curing in 20°C IPA for 1 min, it becomes sufficiently rigid to penetrate porcine muscle tissue with a force (8.2 ± 0.6 N), comparable to that of commercial steel needles (Figure 6d; Movies S5 and S6) [45]. Once inserted a simulated physiological environment (37°C normal saline), the hydrogel needle softens within 30 s, becoming compliant with surrounding tissues to minimize the risk of microtrauma, while still allowing effective fluid delivery (Figure 6e; Movie S7). Cell viability assay results are summarized in Figure S7 and Table S1, confirming the cytocompatibility of the material. This rapid soft-rigid-soft transition, illustrated conceptually in Figure 6c, provides a promising strategy for safer medical devices that mitigate needlestick injuries and tissue damage.

3 | Conclusion

In summary, we have developed a PAA-SF composite hydrogel by blending SF with PAA in a defined ratio. By systematically tuning the SF content within the PAA network, the resulting hydrogels exhibit tunable shape recovery kinetics across multiple programming environments, along with enhanced mechanical properties and strong adhesion. Leveraging these attributes, including spatiotemporal control of shape memory and excellent cytocompatibility, we designed two representative smart devices: a four-jaw gripper capable of object retrieval in confined spaces and a hydrogel needle that reversibly switches between soft and rigid states. This work offers valuable insights into the integration of SF-based shape memory polymers into biomedical applications. Moreover, the ability to program shape recovery in diverse environments provides unprecedented control over the deformation behavior of SMPs, paving the way for advanced SMP-based devices in soft robotics, minimally invasive medicine, and beyond.

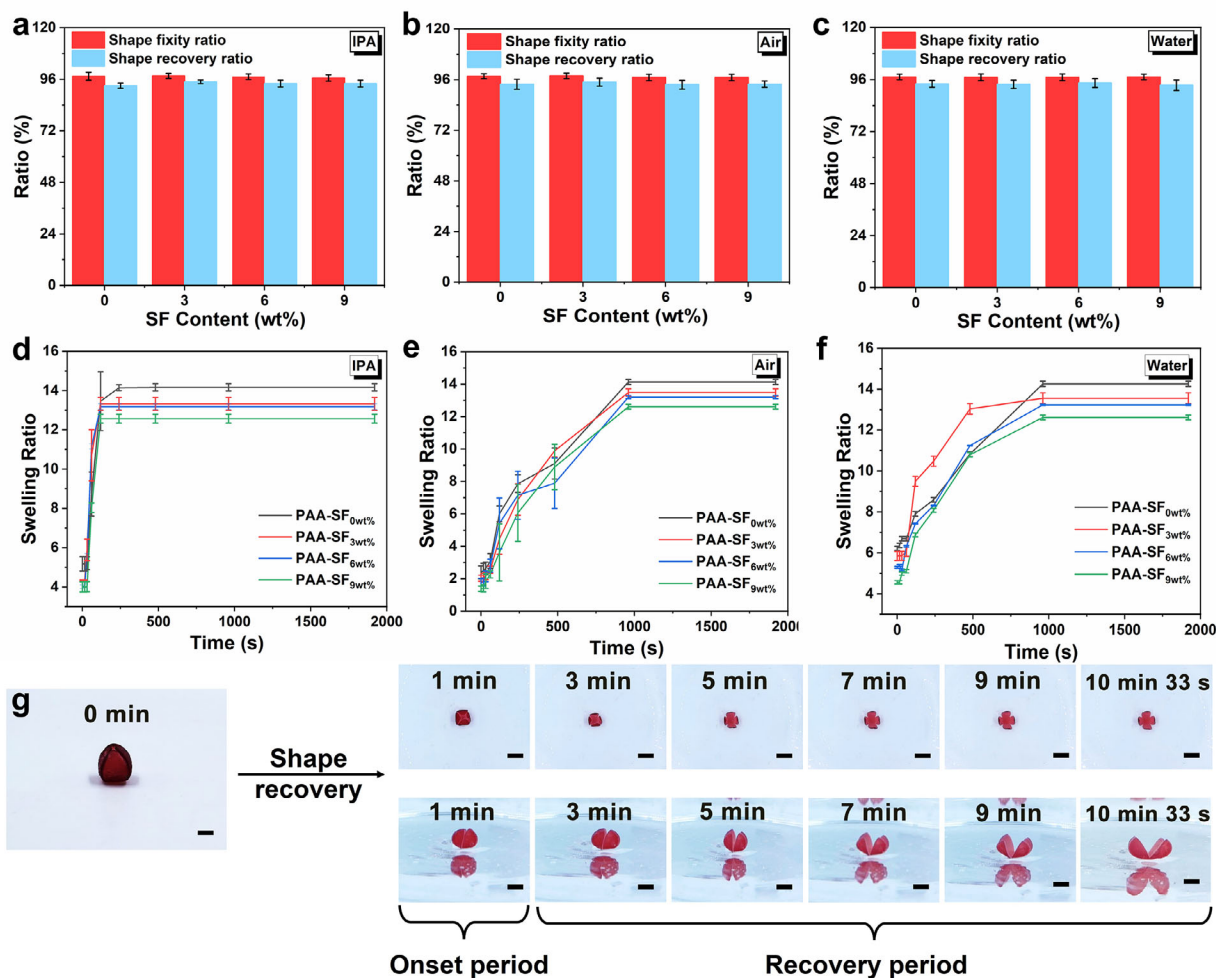


FIGURE 4 | (a–c) R_f and R_r of PAA-SF hydrogels programmed in 20°C IPA, 80°C air, and 80°C water for 5 min, respectively. (d–f) Swelling kinetics of PAA-SF hydrogels with varying SF content programmed in 20°C IPA, 80°C air, and 80°C water for 5 min, respectively. (g) Photographs illustrating the recovery process of a flower-shaped PAA-SF_{6 wt.%} hydrogel. All scale bars are 10 mm. All the error bars correspond to s.d. ($n = 5$).

4 | Experimental Section

4.1 | Materials

Bombyx mori cocoons were obtained from the silkworm gene bank (Southwest University, Chongqing, China). Sodium carbonate (Na_2CO_3), Lithium bromide (LiBr), cellulose dialysis membranes (Spectra/Por1, Biotech Cellulose Ester membranes, MWCO: 3500, Spectrum Laboratories Inc.) were purchased from Sigma-Aldrich, Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China), Sinopharm Chemical Reagent Co., Ltd (Shanghai, China), and Dow Corning, respectively. Acrylic acid (AAc), $\text{N,N}'$ -methylenebisacrylamide (BIS), Lithium phenyl-2,4,6-trimethylbenzoylphosphinate (LAP), and calcium acetate (CaAs), were purchased from Aladdin. Sudan III and crystal violet were purchased from Macklin.

4.2 | Preparation of Silk Fibroin Solution

An aqueous Na_2CO_3 solution (2.12 g L^{-1} , 2800 mL) was prepared and heated to boiling. 7 g of raw silkworm cocoons were cut into small pieces (approximately the size of a 1-yuan coin) and

added to the boiling Na_2CO_3 solution. The mixture was boiled and stirred for 30 min to dissolve sericin. After degumming, the degumming part was collected and thoroughly rinsed with deionized water to remove residual sericin, obtaining purified silk fibroin, which was then dried in the dark. After this, dried silk fibroin (3 g) was dissolved in LiBr solution (0.8 g mL^{-1}). After 1 h of dissolution, the solution turned amber in color. After 3 h, the solution was transferred into a dialysis bag (with a molecular weight of 10 kDa) and dialyzed in deionized water for 3 days, with frequent water changes. After the end, the dialysis bag solution was removed and centrifuged at 5000 rpm to obtain pure silk fibroin solution. Ultimately, the silk fibroin was obtained in lyophilized powder form via freeze-drying at -80°C .

4.3 | Preparation of Photocurable Hydrogel Precursor

AAc (16 g), 0.25 mol.% BIS (relative to the concentration of AAc), LAP (0.2 g), and deionized water (4 mL) were mixed to form PAA hydrogel precursor. PAA-SF hydrogel precursors containing 3, 6, and 9 wt.% SF (relative to the

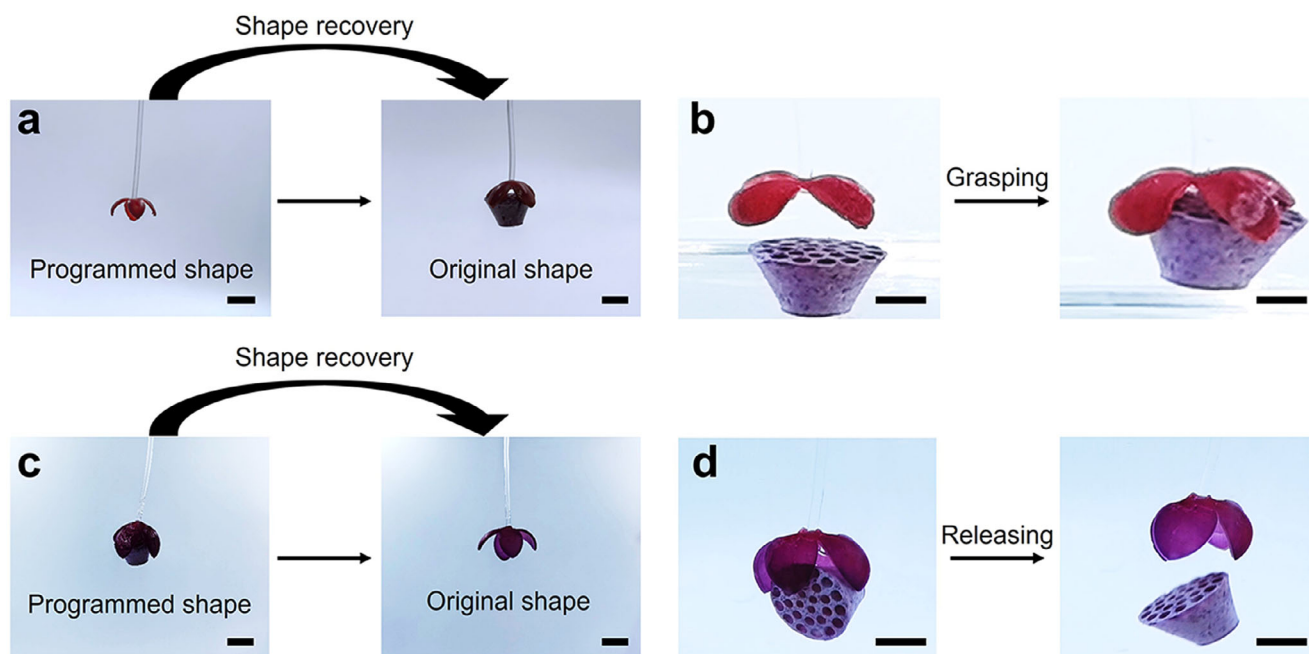


FIGURE 5 | Smart four-jaw gripper. (a) Photograph of 3D-printed PAA-SF₆ wt.% four-jaw gripper with a closed original configuration. (b) Snapshots of the gripper approaching an object (left) and successfully grasping it (right). (c) Photograph of 3D-printed PAA-SF₆ wt.% four-jaw gripper with an open original configuration. (d) Snapshots of the gripper approaching an object before release (left) and after complete release (right). All scale bars are 10 mm.

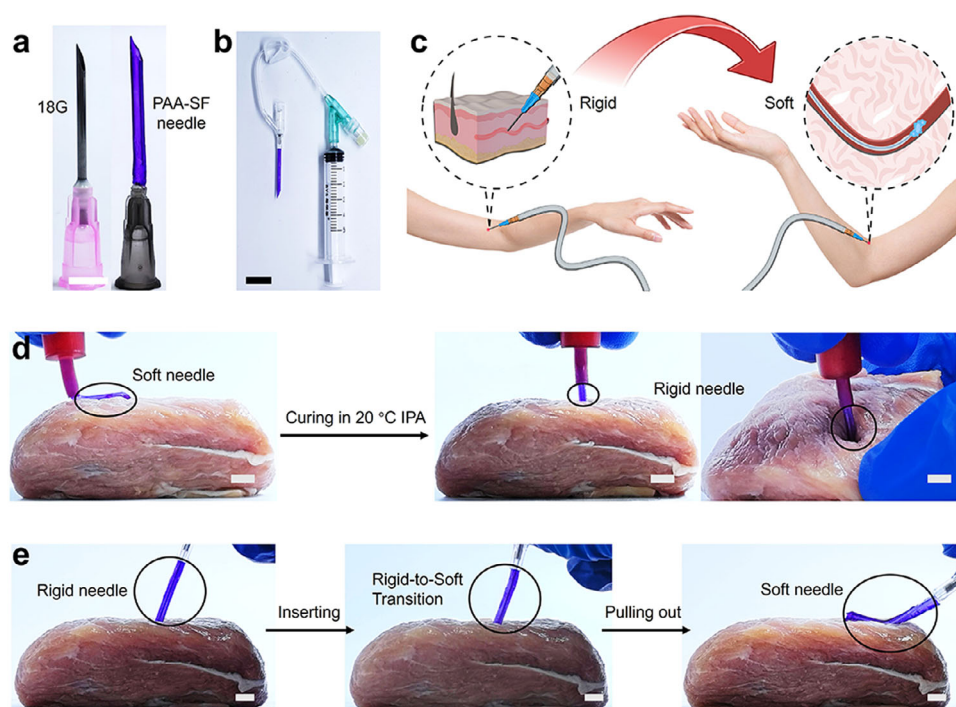


FIGURE 6 | Stiffness-tunable hydrogel needle. (a) Photograph of a commercial 18G needle (left) and the PAA-SF₆ wt.% needle (right). (b) Photograph showing the PAA-SF₆ wt.% needle attached to a 5 mL syringe. (c) Conceptual illustration of the key features of the stiffness-tunable PAA-SF₆ wt.% needle. (d) Photographs of the PAA-SF₆ wt.% needle inserting porcine muscle tissue before (left) and after (right) curing. (e) The PAA-SF₆ wt.% needle during a simulated syringe injection procedure for drug delivery. All scale bars are 10 mm.

concentration of AAc) were prepared by dispersing the lyophilized SF powder into PAA precursor, followed by ultrasonication to obtain a uniform dispersion. For enhanced visualization, 0.001% Sudan III and 1 mg mL⁻¹ crystal violet solution were optionally added.

4.4 | Synthesis of Shape Memory Hydrogels

Two glass plates and a silicon rubber spacer were combined to form a mold. Spacer thicknesses of 0.5, 1.5, and 3.0 mm were used. The PAA and PAA-SF precursors were poured into the mold and cured in a UV curing box with output power of 80 W for 1 min. The cured film was then immersed in 1 M calcium acetate solution for 3 days, followed by immersion in deionized water for an additional 3 days, to obtain the shape memory hydrogels.

4.5 | Characterization of Shape Memory Function

Shape memory properties were quantitatively evaluated using rectangular hydrogel strips (45 mm × 15 mm × 1.5 mm). Samples were first deformed into a U-shaped temporary configuration and then fixed in one of three programming environments (20°C IPA, 80°C air, or 80°C water) for 5 min. After programming, the samples were immersed in 20°C water to initiate shape recovery. The entire process was recorded, and the bending angle (θ) was tracked frame-by-frame. The onset time (t_{onset}) was defined as the time required to reach 10% of the total angular recovery, while the recovery time (t_{recovery}) was defined as the time to reach 95% recovery, and timing begins upon immersion of the samples into the recovery medium (20°C water). The shape fixity ratio (R_f) and shape recovery ratio (R_r) were calculated using standard Equations 1 and 2.

$$R_f = (\theta_{\text{fixed}} / \theta_{\text{deformed}}) \times 100\% \quad (1)$$

$$R_r = (\theta_{\text{deformed}} - \theta_{\text{recovered}}) / \theta_{\text{deformed}} \times 100\% \quad (2)$$

4.6 | 3D Printing

Models of four-jaw grippers and the hydrogel needle were designed by SolidWorks 2025. A custom, top-down digital light processing printer (nanoArch P150) was used for 3D printing of SF-based hydrogels. The printer operated in a configuration with a layer thickness of 100 μm and a lateral resolution of up to 700 μm . Specific printing parameters are summarized in Tables S2 and S3. After printing, the 3D objects were soaked in 1 M calcium acetate solution for 3 days and then in deionized water for 3 days to obtain the 4D-printed hydrogel.

4.7 | Fourier Transform Infrared Spectroscopy (FTIR)

PAA-SF hydrogels with different SF contents were cut into small pieces and quickly frozen in liquid nitrogen before freeze-drying at -80°C . The dried samples were ground into powder. FTIR

spectra were recorded using a Nicolet iS20 FTIR spectrometer (Thermo Fisher Scientific Nicolet iS20, USA).

4.8 | Variable Temperature Fourier Transform Infrared Spectroscopy (VT-FTIR)

PAA-SF hydrogels with 0, 6, and 9 wt.% SF were cut into small pieces, rapidly frozen in liquid nitrogen, and freeze-dried at -80°C . The dried samples were ground into powder. VT-FTIR measurements commenced at 20°C , followed by a temperature ramp of $10^\circ\text{C min}^{-1}$ to 40°C , 60°C , and 80°C . At each target temperature, samples were equilibrated for 5 min before spectral acquisition, yielding four spectra for the heating phase. Upon completion of the 80°C measurement, room-temperature deionized water was added to fully submerge the powder, maintaining this condition throughout the subsequent cooling phase. The sample was then allowed to cool naturally to approximately 20°C over 20 min without active temperature control. During this period, spectra were collected every 30 s, and the sample temperature was monitored. All VT-FTIR spectra were recorded using a Nicolet iS20 FTIR spectrometer (Bruker VERTEX 80v, Germany).

4.9 | Raman Spectroscopy

Raman spectra were obtained using a confocal micro-Raman spectrometer (Horiba LabRAM HR Evolution, Japan) equipped with a 785 nm laser. The spectra of the sample fragments were collected at 25°C in the range of $400\text{--}1800\text{ cm}^{-1}$.

4.10 | Scanning Electron Microscopy (SEM) Observation

Hydrogel samples with SF contents of 0, 3, 6, and 9 wt.% were first frozen in liquid nitrogen and then freeze-dried at -80°C . A Hitachi S-4800 SEM operating at 10 kV was employed for morphological observation. Furthermore, PAA-SF_{6 wt.%} hydrogels were programmed in three different environments (20°C IPA, 80°C air, and 80°C water) for 5 min to study the resulting morphological changes. After programming, these samples were immediately quenched in liquid nitrogen, freeze-dried, and subsequently imaged by SEM. The observed porous structure originates from water sublimation during freeze-drying, wherein the pores represent the original water domains in the phase-separated hydrogel and thus directly reflect the degree of phase separation.

4.11 | Mechanical Properties

The mechanical properties of the PAA-SF hydrogels with different SF contents were measured using an AGS-X electronic tensile machine (Shimadzu, Japan) at a deformation rate of 5 mm min^{-1} in air at room temperature. Each test was repeated at least five times under identical environmental temperature and humidity. Prior to testing, hydrogel samples were trimmed to a rectangular shape ($35\text{ mm} \times 10\text{ mm} \times 3.0\text{ mm}$). Tests were terminated upon specimen fracture.

4.12 | Swelling Kinetics Measurement

After programming, the surfaces of each hydrogel sample were blotted dry and immediately weighed as W_p . Samples were then immersed in excess deionized water at room temperature (20°C). At designated time points, samples were removed, surface moisture was absorbed, and they were weighed as W_t before returning to the water. This process continued until constant weight indicated swelling equilibrium. Finally, all samples were freeze-dried to determine the dry weight as W_d . The swelling ratio at each time point was calculated by equation (3).

$$\text{Swelling ratio} = (W_t - W_d) / W_d \quad (3)$$

4.13 | Water Swelling Performances of SF-Based Hydrogels

Each hydrogel sample was soaked in deionized water at room temperature (20°C) to achieve their equilibrium swelling. The swollen weight was recorded as W_s . The fully swollen hydrogels were then freeze-dried and weighed as W_d . Five parallel samples were measured simultaneously, and the equilibrium water content was calculated according to equation (4).

$$\text{Equilibrium water content} = (W_s - W_d) / W_d \quad (4)$$

4.14 | Adhesion Properties Measurement

The adhesiveness of hydrogels was evaluated using AGS-X electronic tensile machine (Shimadzu, Japan) at room temperature with a speed of 50 mm min⁻¹. Pig skin was used as substrates. Hydrogel specimens (30 mm × 12 mm × 2.0 mm) were sandwiched between two pig-skin substrates and pressed to ensure intimate contact. Lap-shear tests were performed until complete detachment of the hydrogel from either substrate.

4.15 | Elaboration on the Specific Infusion Procedure

A container was filled with normal saline and placed in a thermostatic water bath set to 37°C [46]. De-skinned porcine muscle tissue was fully submerged in the 37°C water bath and equilibrated for at least 30 min to ensure that the internal temperature of the tissue reached approximately 37°C. The needle penetration and subsequent fluid injection tests (Figure 6d,e; Movie S7) were conducted with the tissue remaining submerged in the 37°C water bath. This setup simulated a physiological fluid environment and provided a more biologically relevant context for evaluating needle performance.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adfm73616-sup-0001-SupMat.docx **Supporting File:** adfm73616-sup-0002-MovieS1-S7.zip